

- Vector Spaces.

Let K be a field (Think of $K = \mathbb{R}$ or $K = \mathbb{C}$)

A vector space V over K is a set with:

$$\begin{aligned} +: V \times V &\rightarrow V & (\text{addition}) \\ \cdot: K \times V &\rightarrow V & (\text{scalar multiplication}) \end{aligned}$$

closed in V

such that: $u, v, w \in V, \lambda, \mu \in K$

- (1). $(u+v)+w = u+(v+w)$.
- (2). $u+v = v+u$
- (3). $\exists 0_V \in V: u+0_V = 0_V+u = u$
- (4). $\forall u \in V, \exists -u \in V: u+(-u) = (-u)+u = 0_V$
- (5). $(\lambda\mu)u = \lambda(\mu u)$
- (6). $(1_K)u = 1_K \cdot u = u$
- (7). $\lambda(u+v) = \lambda u + \lambda v$
- (8). $(\mu+\lambda)u = \mu u + \lambda u$

- What is a vector?

- "Vector is something with size and a direction",



- Still, having in mind is useful.

• Examples of vec. spaces.

$$K^n = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_i \in K, i=1,2,\dots,n \right\}$$

$$K_n = \{ (x_1, x_2, \dots, x_n) : x_i \in K, i=1,2,\dots,n \}$$

$$K^{m \times n} = \{ (a_{ij})_{m \times n} : a_{ij} \in K, i=1,\dots,m, j=1,\dots,n \}$$

$$K[x] = \left\{ a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i : \begin{array}{l} a_i \in K, n \in \mathbb{N}, j=0,\dots,n \\ \text{Convention: The coefficient before the highest degree is non-zero} \end{array} \right\}$$

- Vector subspace $W \subseteq V$ if

$$w_1, w_2 \in W, \mu, \lambda \in K$$

we have $\mu w_1 + \lambda w_2 \in W$. Denote as $W \subseteq V$

$(V, +, \cdot)$ repeatedly perform these operation:

$$\text{Linear combination: } v = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = \sum_{i=1}^n \lambda_i v_i$$

v is a linear combination of $v_1, v_2, \dots, v_n$

- Given $v_1, v_2, \dots, v_n \in V$ vary the coefficients $\lambda_1, \dots, \lambda_n \in K$ to get

$$\text{Span}(v_1, v_2, \dots, v_n) = \left\{ \sum_{i=1}^n \lambda_i v_i : \lambda_i \in K, i=1,\dots,n \right\}$$

If $V = \text{Span}(v_1, \dots, v_n)$, we say V is generated by $v_1, \dots, v_n$ or $v_1, \dots, v_n \text{ span } V$.

- Easy to see: $\text{Span}(v_1, \dots, v_n)$ is a subspace of V

- Proposition: "Span is the smallest subspace."

Take $v_1, \dots, v_n \in V$ then:

$$\text{Span}(v_1, \dots, v_n) = \bigcap_{\substack{W \subseteq V \\ v_1, \dots, v_n \in W}} W$$

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Equivalently if $W \subseteq V$ and $v_1, \dots, v_n \in W$, then

$$\text{Span}(v_1, \dots, v_n) \subseteq W$$

Span $(v_1, \dots, v_n) \subseteq W$.
 Proof. We prove set equality by showing " $\subseteq$ " and " $\supseteq$ ".
 Left $\subseteq$ Right. Take $v \in \text{Span}(v_1, \dots, v_n)$. by defn of the span
 $v = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n$

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Since $v_1, \dots, v_n \in W$ and W subspace.

Since $v_1, \dots, v_n \in W$

$$v = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \in W \text{ for every } W$$

$$\Rightarrow \text{Span}(v_1, \dots, v_n) \subseteq \bigcap_{\substack{W \leq V \\ v_1, \dots, v_n \in W}} W$$

Right $\subseteq$ Left. Note $\text{Span}(v_1, \dots, v_n)$ is itself a subspace with $v_1, \dots, v_n \in \text{Span}(v_1, \dots, v_n)$.

This means $\text{Span}(V_1, \dots, V_n)$ is one of the U_i being intersected.

$$\Rightarrow \bigcap_{\substack{W \leq V \\ v_1, \dots, v_n \in W}} W \subseteq \text{Span}(v_1, \dots, v_n)$$

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Linear Independence:

Independence:
 $v_1, \dots, v_n \in V$ is linearly independent if
 $\lambda_1 v_1 + \dots + \lambda_n v_n = 0_V$ then $\lambda_1 = \dots = \lambda_n = 0_K$

Examples: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{R}^2$ Lin. indep.

2) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \end{pmatrix} \in \mathbb{R}^2$ Lin. dep.

Reason: $2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} -2 \\ -2 \end{pmatrix} = 0_{\mathbb{R}^2}$ and $2 \neq 0 \neq 1$.

Prop. (Uniqueness of representation).

$v_1, \dots, v_n \in V$ is linearly indep. iff " $\Leftrightarrow$ "

every vector $v \in \text{Span}(v_1, \dots, v_n)$ has a unique representation as linear combinations of $v_1, \dots, v_n$.

That is $\exists! \lambda_1, \dots, \lambda_n \in \mathbb{K} : v = \sum_{i=1}^n \lambda_i v_i$.

Proof: ($\Rightarrow$). Suppose $v_1, \dots, v_n$ lin. indep.

and that $v = \lambda_1 v_1 + \dots + \lambda_n v_n$

$$= \mu_1 V_1 + \dots + \mu_n V_n$$

$$\Rightarrow (\lambda_1 - \mu_1) v_1 + \dots + (\lambda_n - \mu_n) v_n = 0_v$$

By linear indep. $\lambda_i - \mu_i = 0$ $\forall i=1, \dots, n$.

Then $\lambda_i = \mu_i \quad \forall i \Rightarrow$ Unique.

($\Leftarrow$). Suppose unique representation, then

In particular:

$$\Rightarrow O_r = O_{1K} \cdot V_1 + \dots + O_{iK} V_n. \}$$

$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(x_k) = \int_a^b f(x) dx$ holds for a continuous

In particular...
 $\rightarrow O_V = 0_K V_1 + \dots + 0_K V_n$
 $\rightarrow$ Suppose $O_V = \lambda_1 V_1 + \dots + \lambda_n V_n$ holds for a nontrivial set of $\lambda_1, \dots, \lambda_n$
 Then O_V would have 2 different rep.
 Contradiction!
 $\square$

Prop: $V_1, \dots, V_n \in V$ is lin. dep. iff for some $1 \leq i \leq n$,
 $V_i \in \text{Span}(V_1, \dots, V_{i-1}, V_{i+1}, \dots, V_n)$.
 missing V_i

Proof: ($\Rightarrow$). Suppose $V_1, \dots, V_n$ lin. dep.

$$O_V = \lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_n V_n$$

$\lambda_1, \dots, \lambda_n$ are not all zeros.

Take $\lambda_i \neq 0$ because K is a field, $\lambda_i^{-1} \in K$

$$O_V = \frac{\lambda_1}{\lambda_i} V_1 + \frac{\lambda_2}{\lambda_i} V_2 + \dots + \underbrace{\frac{\lambda_i}{\lambda_i}}_{1_K} V_i + \dots + \frac{\lambda_n}{\lambda_i} V_n$$

$$\Downarrow$$

$$V_i = - \sum_{j \neq i} \frac{\lambda_j}{\lambda_i} V_j \Rightarrow V_i \in \text{Span}(V_1, \dots, V_{i-1}, V_{i+1}, \dots, V_n)$$

($\Leftarrow$). Suppose $V_i \in \text{Span}(V_1, \dots, V_{i-1}, V_{i+1}, \dots, V_n)$.

Then $V_i = \sum_{j \neq i} \lambda_j V_j$. then

$$O_V = 1_K V_i - \sum_{j \neq i} \lambda_j V_j \Rightarrow \text{lin. dep.}$$

$\square$

- Basis: Take W (sub)space. A basis of W is a set of linear indep vectors that spans W .
 That is $\underline{V_1, \dots, V_n} \in W$. lin. indep. and
 $W = \text{Span}(V_1, \dots, V_n)$.

Note: In this course we concern only vector spaces with a finite basis.

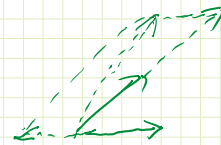
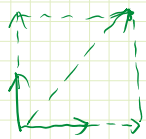
Fact: Basis exists for any vector space (No restriction on dimension or not)

Fact: Basis exists for any vector space (irrespective of whether the basis is finite or not)

We will not prove this. It uses Zorn's Lemma.

Fact: Two bases for a vector space have the same number of elements for finite bases.

Example: $\mathbb{R}^2 = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$



We call the number of elements in these finite bases the dimension of W , denoted as $n = \dim W$.

- Understanding Basis.

1. Basis is the minimal spanning set.

$v_1, \dots, v_n$ a basis for W . Take out any v_i .
 $\text{Span}(v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n) \subsetneq W$

2. Basis is the maximal lin. indep. set.

add any $v \in W$ to $v_1, \dots, v_n$ (basis) then $v, v_1, \dots, v_n$ is lin. dep.

- Change of Basis.

Take a basis $\mathcal{B} = (v_1, \dots, v_n)$ for V .

For every $x \in V$, $x = x_1 v_1 + \dots + x_n v_n$

Take
For every $x \in V$, $x = x_1 v_1 + \dots + x_n v_n$
(Notation convention) $= (v_1, \dots, v_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$.

The vector $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ is called the coordinate vector for x .

Now take $\mathcal{E} = (e_1, \dots, e_n)$, $\mathcal{F} = (f_1, \dots, f_n)$ two bases for V . Because each $f_i \in V$.

$$f_1 = p_{11}e_1 + p_{21}e_2 + \dots + p_{n1}e_n$$

$$f_2 = p_{12}e_1 + p_{22}e_2 + \dots + p_{n2}e_n$$

$$\vdots$$

$$f_n = p_{1n}e_1 + p_{2n}e_2 + \dots + p_{nn}e_n$$

Define $P = \begin{pmatrix} p_{11} & p_{12} & \dots & p_{1n} \\ p_{21} & p_{22} & \dots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \dots & p_{nn} \end{pmatrix}$ then we can write

$$(f_1, \dots, f_n) = (e_1, \dots, e_n) P = (e_1, \dots, e_n) \begin{pmatrix} p_{11} & p_{12} & \dots & p_{1n} \\ p_{21} & p_{22} & \dots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \dots & p_{nn} \end{pmatrix}$$

P is called the transition matrix from $\mathcal{E}$ to $\mathcal{F}$.

The i th column of P is the coordinate vector of $f_i \in \mathcal{F}$ under basis $\mathcal{E}$.

Claim: P is invertible.

$(f_1, \dots, f_n) = (e_1, \dots, e_n) P$ and

Claim: P is invertible.

Proof: Suppose $(f_1, \dots, f_n) = (e_1, \dots, e_n) P$ (*) and

$$(e_1, \dots, e_n) = (f_1, \dots, f_n) Q \quad (**)$$

Then $(e_1, \dots, e_n) = \underbrace{(f_1, \dots, f_n)}_{\text{by (*)}} Q \quad \text{by (**)}$

$$= (e_1, \dots, e_n) P Q$$

but $(e_1, \dots, e_n) = (e_1, \dots, e_n) I_n$

Therefore $PQ = I_n$.

Starting from $(f_1, \dots, f_n) = \underbrace{(e_1, \dots, e_n)}_{\substack{\parallel \\ \downarrow}} P \quad \text{by (*)}$

$$(f_1, \dots, f_n) I_n = (f_1, \dots, f_n) Q P \quad \text{by (**)}$$

Then $QP = I_n$. Thus $Q = P^{-1}$ $\square$

Now Let's look at the coordinate vectors for $x \in V$ under $E = (e_1, \dots, e_n)$ and $F = (f_1, \dots, f_n)$.

$$x = x_1 e_1 + \dots + x_n e_n = y_1 f_1 + \dots + y_n f_n$$

$$(e_1, \dots, e_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (f_1, \dots, f_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

(Since from change of basis $(f_1, \dots, f_n) = (e_1, \dots, e_n) P$)

$$(e_1, \dots, e_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (e_1, \dots, e_n) P \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$\parallel$ $\parallel$
 X Y

We get $X = PY$ change of basis for
coordinate vectors

Linear Maps and Their Matrices.

- What is a linear maps? over $K \rightarrow \dots$

Linear Maps and Their Matrices.

- What is a linear map? over K

Let V, W be two vec spaces. $T: V \rightarrow W$

is a linear map if $u, v \in V, \lambda, \mu \in K$

$$T(\lambda u + \mu v) = \lambda T(u) + \mu T(v).$$

If $W=V$ this called a linear transformation or a linear operator.

Lin. maps between V and $W: \mathcal{L}(V, W) \cong$

Lin. transformations from V to $V: \mathcal{L}(V)$.

- We can define $+$: $(T+S)(u) = T(u) + S(u)$ } Defining
 $\cdot$: $(\lambda T)(u) = \lambda T(u)$ } $T+S$
 $T, S \in \mathcal{L}(V, W), \lambda \in K$ } λT

Check: $T+S \in \mathcal{L}(V, W)$ and $\lambda T \in \mathcal{L}(V, W)$.

Then $\mathcal{L}(V, W)$ is a vec. space over K .

Take $W=V$, $\mathcal{L}(V)$ is also a vec. space over K .

- What do we need to know to understand the behaviour of a linear map T ?

Ans: Look at what it does to the basis vectors of the domain of T .

Take $T \in \mathcal{L}(V, W)$ and $E = (e_1, \dots, e_n)$ a basis for the domain V .

For any $v \in V$ $v = x_1 e_1 + x_2 e_2 + \dots + x_n e_n$. (*)

Apply T to both sides of (*), by linearity

$$\begin{aligned} T(v) &= T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) \\ &= x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n). \end{aligned}$$

So once we know $T(e_1), \dots, T(e_n)$, we can recover what T does to any vector $v \in V$.

Live in codomain W $\left\{ \begin{array}{l} T(e_1) = a_{11}f_1 + a_{12}f_2 + \dots + a_{1n}f_n \\ T(e_2) = a_{21}f_1 + a_{22}f_2 + \dots + a_{2n}f_n \\ \vdots \\ T(e_n) = a_{n1}f_1 + a_{n2}f_2 + \dots + a_{nn}f_n \end{array} \right.$ $\dim V = n$
 Take $\mathcal{F} = (f_1, \dots, f_m)$ a basis of W $\left\{ \begin{array}{l} \vdots \\ \vdots \end{array} \right.$ $\dim W = m$

$$\begin{aligned} T(e_1, e_2, \dots, e_n) &= (f_1, f_2, \dots, f_m) \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \\ &\quad \parallel \\ &= (f_1, f_2, \dots, f_m) A \end{aligned}$$

A is called the matrix of T under E and $\mathcal{F}$.

$\parallel$ the coordinate vector

A is called the matrix of T under $\mathcal{E}$ and $\mathcal{F}$.

The i^{th} column of A is the coordinate vector of $T(e_i)$ under basis $\mathcal{F}$ (in W).

- Let's look at coordinate vectors for any $v \in V$

$$v = x_1 e_1 + x_2 e_2 + \dots + x_n e_n = (e_1, \dots, e_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad (*)$$

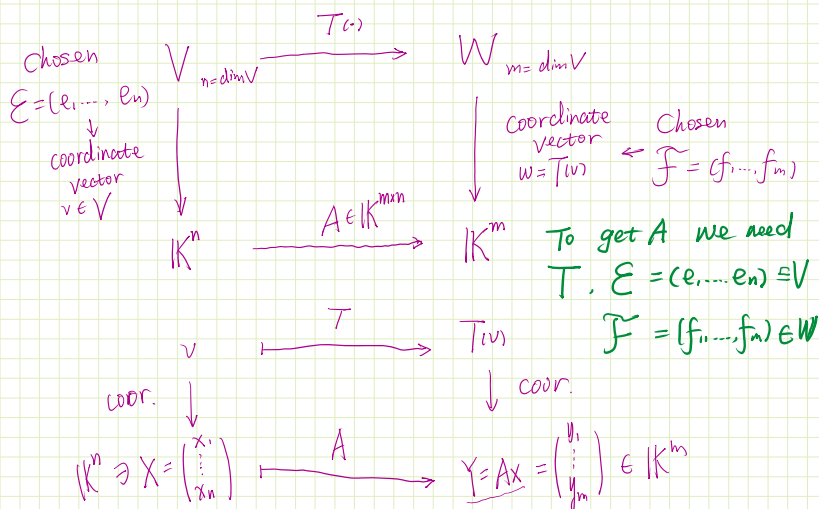
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Apply T to both sides of $(*)$.

$$T(v) = T(e_1, \dots, e_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (f_1, \dots, f_m) A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$= (f_1, \dots, f_m) AX.$$

$$\left(\begin{matrix} A_{mn} & X_{n \times 1} = Y \\ n = \dim V & m = \dim W \end{matrix} \right)$$



- Vec. space & subspace
- Linear comb. & Span
- Linear (in) dep.
- Basis & change of Basis
- Linear maps and their matrices

Coming Next: Rank...